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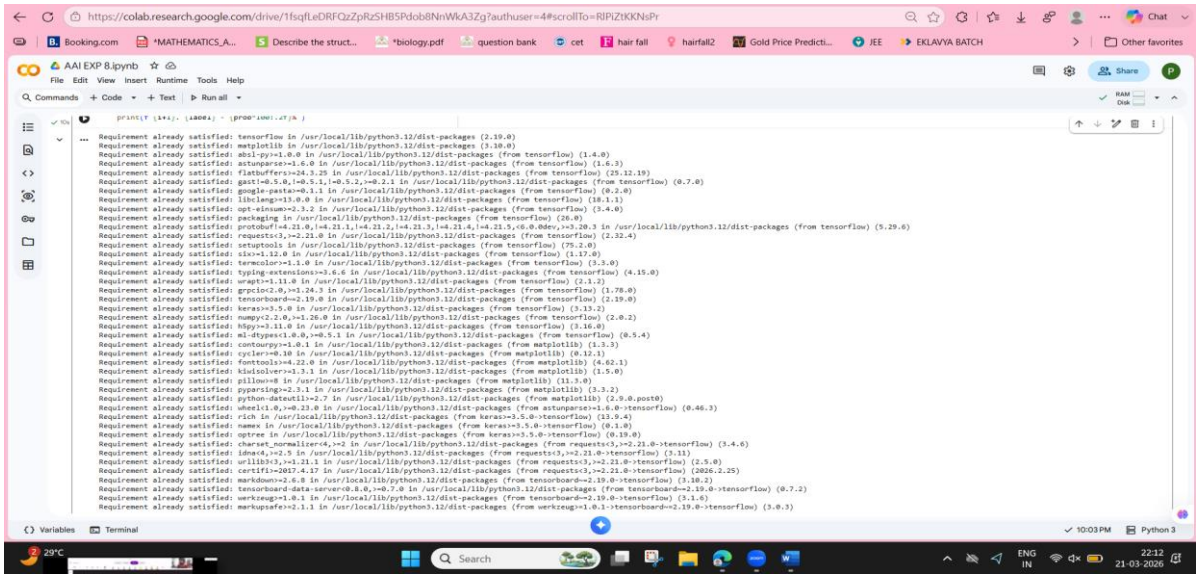
AAI EXPERIMENT NO : 08

CODE:

```
# Install dependencies (if needed)

!pip install tensorflow matplotlib

import tensorflow as tf

import numpy as np

import matplotlib.pyplot as plt

from tensorflow.keras.applications import MobileNetV2

from tensorflow.keras.applications.mobilenet_v2 import preprocess_input,
decode_predictions

from tensorflow.keras.preprocessing import image

# Load Pre-trained Model

model = MobileNetV2(weights='imagenet')

# Download Sample Image

img_path = tf.keras.utils.get_file(

    'sample.jpg',

    'https://storage.googleapis.com/download.tensorflow.org/example_images/grace_hopper.jp
g' # Changed to a different reliable URL

)

# Load and Preprocess Image

img = image.load_img(img_path, target_size=(224, 224))

img_array = image.img_to_array(img)

img_array = np.expand_dims(img_array, axis=0)

img_array = preprocess_input(img_array)

# Make Prediction

predictions = model.predict(img_array)

# Decode Predictions

decoded = decode_predictions(predictions, top=3)[0]

# Display Image

plt.imshow(img)
```

```
plt.axis('off')
plt.title("Input Image")
plt.show()
# Print Results
print("Top Predictions:")
for i, (imagenetID, label, prob) in enumerate(decoded):
    print(f"{i+1}. {label} - {prob*100:.2f}%")
```

OUTPUT:



```
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich->keras>=3.5.0->tensorflow) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich->keras>=3.5.0->tensorflow) (0.1.2)
Downloading data from https://storage.googleapis.com/download.tensorflow.org/example_images/grace_hopper.jpg
61306/61306 ━━━━━━━━━━━━━━━━━━━━ 0s 1us/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 2s 2s/step
Downloading data from https://storage.googleapis.com/download.tensorflow.org/data/imagenet_class_index.json
35363/35363 ━━━━━━━━━━━━━━━━━━━━ 0s 1us/step
```

Input Image

```
Top Predictions:
1. military_uniform - 88.23%
2. suit - 1.33%
3. Windsor_tie - 0.61%
```